

Name: \_\_\_\_\_ ID: \_\_\_\_\_

Section: L4

Q. 1: [2 + 6 + 2 = 10 points] A team from Birthday Party Company of Karachi has asked you to **design** and **implement** a program for a **musical chairs** game for kids. The game steps are as follows:

Given an initial arrangement of  $n$  children, and  $n$  chairs, there will be  $n - 1$  rounds of the game:

1. In each round, the children move  $s$  steps to the right while a song of  $s$  steps is being played. Here each step corresponds to moving from one chair to another.
2. After  $s$  steps the song stops, the rightmost chair is removed and the child on that chair is 'out'.
3. At the end of  $n - 1$  rounds there is only 1 chair left and 1 child. That child is the 'winner'.

They have further explained it to you using the following example.

Let  $n = 5$ ,  $s = 2$ , i.e. there are 5 children and a song of length 2 is played. Let's name the children as:



The following table gives steps of the game (Note that  $n = 5$ ,  $s = 2$ ):

| Round | Arrangement of Children before the round    | Arrangement of Children after the round              |
|-------|---------------------------------------------|------------------------------------------------------|
| 1     | <br>Ali      Eva      Naz      Sam      Tim | <br>Sam      Tim      Ali      Eva      Naz          |
| 2     | <br>Sam      Tim      Ali      Eva          | <br>Ali      Eva      Sam      Tim                   |
| 3     | <br>Ali      Eva      Sam                   | <br>Eva      Sam      Ali                            |
| 4     | <br>Eva      Sam                            | No more rounds,<br>Eva      Sam Eva is the<br>winner |

You plan to implement the solution using a deque. Suppose you have a Deque (double ended queue) with the operations allowed on the Deque as follows (all operations take constant time, i.e.,  $O(1)$ ).

- `void addToFront(x)`, `void addToRear(x)` //No element returned.
- `T deleteFromFront()`, `T deleteFromRear()` //Deleted element is returned.
- `size()` //returns the **current** size of the deque as the number of elements.

(a) Given an initial arrangement of children [Dua, Gul, Nur, Zak] with  $s = 3$ , what is the arrangement of children for after each round? Who is the winner?

| Initially            | After Round 1 | After Round 2 | After Round 3 |
|----------------------|---------------|---------------|---------------|
| [Dua, Gul, Nur, Zak] |               |               | Winner        |

- (b) Write an algorithm (as a method) in pseudocode that simulates a game of musical chairs. It takes in as input a deque that has the names of  $n$  children. After each round, where each round consists 's' number of steps, the algorithm finds and prints the resulting arrangement of children. Finally, the algorithm prints the name of the winner. The method signature is:

**void play\_musical\_chairs(int s, ArrayDeque<String> kids)**

Note: **Your algorithm should use the given deque operations (and variables) if needed.**

To print, use `print(kids)` statement.

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- (c) What is the time complexity of your algorithm in terms of the number of deque operations. Assume `song_length = number of steps = s`, `number of children = n`. Give reason(s) for your answer.

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